Class 4

Science

Solutions

Olympiad Program

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1. Plant around us

Worksheet – 1

1. **Correct answer – B**
   Roots anchor the plant firmly in the soil while root hairs maximize the surface area for water and nutrient absorption.
2. **Correct answer – B**
   Stem, leaves, flowers, and fruits- The shoot system consists of all aerial parts of the plant that grow above the ground.
3. **Correct answer – B**
   Roots are the specialized organs that interface with the soil to uptake essential water and dissolved minerals.
4. **Correct answer – C**
   Conduction of water and minerals via xylem and translocation of food via phloem- The stem acts as a highway, using vascular bundles to move resources between the roots and the leaves.
5. **Correct answer – C**
   Leaves contain chlorophyll and are the primary site where solar energy is converted into chemical energy (food).
6. **Correct answer – B**
   Root hairs are lateral extensions of the root's epidermal cells specifically designed to increase absorption efficiency.
7. **Correct answer – B**
   A fruit is a mature ovary that encloses and protects the seeds until they are ready for dispersal.
8. **Correct answer – B**
   It protects the apical meristem and facilitates downward growth by perceiving gravitational signals- The root cap acts as a physical shield for the growing tip and directs growth toward gravity (geotropism).
9. **Correct answer – B**
   The stem's vascular system allows for the upward movement of water and the downward/multidirectional distribution of sugars.
10. **Correct answer – A**
    Producing food through photosynthesis- The main biological role of a leaf is to capture sunlight and carbon dioxide to synthesize glucose for the plant.

worksheet – 2

1. **Correct answer – C**
   Photosynthesis- This is the biological process where green plants transform light energy into chemical energy in the form of glucose.
2. **Correct answer – A**
   Carbon dioxide, water, and light energy- These three inputs are the essential raw materials required for the chemical reaction that occurs inside the chloroplasts.
3. **Correct answer – C**
   Leaves are the primary photosynthetic organs because they contain high concentrations of chlorophyll and offer a large surface area.
4. **Correct answer – C**
   Oxygen is produced during the photolysis of water and is released into the atmosphere as a byproduct.

5. **Correct answer – B**
This process helps in cooling the plant and creates a suction force that pulls water and minerals up from the roots.
6. **Correct answer – B**
Stomata are the microscopic pores on the leaf surface that regulate the exit of water vapor into the air.
7. **Correct answer – B**
Wood is a natural organic material derived from the secondary xylem of trees and shrubs.
8. **Correct answer – D**
The cellulose fibers found in wood pulp are the primary raw material used in the industrial manufacturing of paper.
9. **Correct answer – B**
Sugarcane and maize are high-energy crops that store solar energy as sucrose and starch, respectively
10. **Correct answer – B**
Leaves are metabolic factories that synthesize various chemical compounds used by humans for medicines and essential oils.

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worksheet – 3

1. **Correct answer – D**
Desert plants use thick stems for storage, deep roots for retrieval, and reduced leaves to prevent water loss.
2. **Correct answer – B**
Cacti have succulent stems that act as reservoirs to sustain the plant during long periods of drought.
3. **Correct answer – C**
Floating leaves allow aquatic plants to access sunlight and atmospheric gases while remaining on the water's surface.
4. **Correct answer – B**
These air-filled tissues (aerenchyma) provide the buoyancy necessary for the plant to stay afloat.
5. **Correct answer – A**
"Drip tips" and waxy surfaces allow heavy rain to run off quickly, preventing fungal growth and leaf damage.
6. **Correct answer – C**
Without chlorophyll, plants cannot capture light energy to convert carbon dioxide and water into glucose.
7. **Correct answer – C**
This specific genus of fungi is the biological source used to produce the antibiotic penicillin.
8. **Correct answer – B**
Fungi are primary decomposers that return essential minerals to the soil by breaking down organic waste.
9. **Correct answer – B**
Fungi are heterotrophic and obtain nutrition by breaking down and absorbing organic matter from their environment.

10. Correct answer – B

Mushrooms are fungi, which are non-green organisms that lack chlorophyll and do not perform photosynthesis.

COMPETATIVE CORNER

SINGLE OPTION CORRECT TYPE

1. Correct answer – C

They have adaptations like deep roots- Terrestrial plants develop varied root systems to anchor themselves and seek water in different soil depths.

2. Correct answer – B

The leaf is the specialized "food factory" containing the majority of the plant's chloroplasts.

3. Correct answer – B

Internal air pockets decrease the plant's density, allowing it to float effortlessly on the water.

4. Correct answer – B

Since they cannot produce their own energy through light, they must obtain it from organic sources.

5. Correct answer – B

This plant has specialized leaves that snap shut to trap prey to supplement its nutrient intake.

6. Correct answer – D

Roots use the process of osmosis and active transport to take in water and minerals from the soil.

7. Correct answer – A

By storing water in their thick stems- Fleshy stems allow desert plants to maintain a water supply while minimizing surface area to reduce evaporation.

8. Correct answer – B

Fungi play a vital role in ecosystems by recycling nutrients from decaying matter back into the food web.

9. Correct answer – B

These plants live in nutrient-poor soil and get essential minerals (like nitrogen) from animal protein.

10. Correct answer – A

Raising flowers above water ensures they are accessible to land-based or flying pollinators like bees.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct answer – A, B, D

Floating plants have roots that hang freely to absorb nutrients, air spaces for buoyancy, and lightweight bodies to stay on the surface.

12. Correct answer – A, B, D

Carnivorous plants trap and digest prey to get nitrogen because they grow in nutrient-poor soils.

13. Correct answer – A, B, D

Roots anchor the plant, absorb water, and sometimes store food (like in carrots); they do not perform photosynthesis.

14. Correct answer – A, B, D

Flowers are reproductive organs that produce seeds and attract pollinators; they do not absorb water from the soil.

15. Correct answer – A, B, D
Seeds contain an embryo (new plant) and stored food; they require specific conditions to germinate but do not photosynthesize.

ASSERTION AND REASNING TYPE

16. Correct answer – A
Chlorophyll is the pigment required to capture light for photosynthesis; without it, the process cannot occur.

17. Correct answer – A
The fleshy, thick stems of desert plants are specifically adapted for water storage (succulence).

18. Correct answer – A
It is classified as insectivorous because its primary adaptation is trapping and digesting animal matter for nutrients.

19. Correct answer – A
Floating leaves keep the photosynthetic surface exposed to sunlight and atmospheric carbon dioxide.

MATCH THE FOLLOWING TYPE

20. Correct answer – (a)-P, (b)-Q, (c)-S, (d)-R

21. Correct answer – (a)-R, (b)-Q, (c)-S, (d)-P

COMPREHENSHION TYPE

Passage1

22. Correct answer – B
Thick stems to store water

23. Correct answer – C
Soft stems for flexibility

24. Correct answer – B
Breaking down dead matter

Passage2

25. Correct answer – B
By snapping its leaf lobes shut

26. Correct answer – B
To get nitrogen

27. Correct answer – C
Leaves

PREVIOUS YEAR QUESTIONS

1. Correct answer – A
Plant will die- Without leaves, the plant cannot perform photosynthesis to make food, leading to starvation and death.

2. Correct answer – C
It can obtain nutrition from other plants as it cannot prepare its own food
Dodder (Cuscuta) is a parasitic plant that lacks chlorophyll and siphons nutrients from a host plant.

3. Correct answer – D

Hydrilla: Cactus: Pitcher plant- This matches the original relationship: **Aquatic** (Duckweed/Hydrilla): **Desert** (Acacia/Cactus): **Insectivorous** (Flytrap/Pitcher plant).

4. Correct answer – A

Oxygen is the gaseous byproduct of the photosynthetic reaction released into the atmosphere.

5. Correct answer – A

Deciduous plants are defined by their habit of shedding leaves annually (usually autumn/winter) and regrowing them in spring.

2. Animals Around Us

Worksheet -1

1. **Correct answer – B**
   Frogs belong to the class Amphibia, which is a group of vertebrates (animals with a backbone
2. **Correct answer – B**
   Vertebrates have an internal skeleton with a vertebral column, while invertebrates lack this structure.
3. **Correct answer – B**
   This tri-segmented body plan is a fundamental anatomical characteristic shared by all insects.
4. **Correct answer – C**
   Amphibians require moist skin for cutaneous respiration (breathing through the skin) and to prevent dehydration.
5. **Correct answer – A**
   Crocodiles are reptiles characterized by their scaly skin, lung respiration, and the fact that they lay hard-shelled eggs.
6. **Correct answer – A**
   Birds lay eggs (oviparous), while mammals mostly give birth to live young (viviparous) and possess mammary glands to feed them.
7. **Correct answer – B**
   The thorax is the middle segment of an insect's body where the locomotive organs (legs and wings) are attached.
8. **Correct answer – A**
   While adult frogs have lungs, they also perform a significant amount of gas exchange through their highly permeable, moist skin.
9. **Correct answer – B**
   These two features are unique to mammals and are not found in birds, reptiles, amphibians, or fish.
10. **Correct answer – B**
    Gills are specialized organs that allow aquatic animals to extract dissolved oxygen directly from the water.

worksheet – 2

1. **Correct answer – B**
   Adaptations are biological features or behaviors developed over generations that give an organism a better chance at living and reproducing.
2. **Correct answer – C**
   Adaptations are biological features or behaviors developed over generations that give an organism a better chance at living and reproducing.
3. **Correct answer – C**
   Carnivores are organisms that derive their energy and nutrient requirements from a diet consisting mainly or exclusively of animal tissue.
4. **Correct answer – B**
   Migration is usually a seasonal response to changes in food availability or the need for a safer climate to raise young.

5. Correct answer – B
The Monarch Butterfly is famous for its incredible multi-generational migration across North America, covering thousands of miles.

6. Correct answer – C
Herbivores require broad, flat molars to break down the tough cell walls (cellulose) found in plant matter.

7. Correct answer – B
The Arctic Tern migrates between the Arctic and Antarctic to follow the "perpetual summer," ensuring they always have enough food and light.

8. Correct answer – B
Thick fur traps air for insulation, and blubber (fat) provides a deep layer of warmth and an energy reserve for animals in freezing temperatures.

9. Correct answer – C
Omnivores are generalist feeders, which allows them to survive in various environments by eating whatever food is available.

10. Correct answer – C
Because plants are difficult to digest, herbivores must grind them into a fine paste to maximize the extraction of nutrients during digestion.

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worksheet 3

1. Correct answer – D
Hens are oviparous animals that lay hard-shelled eggs from which their young hatch.

2. Correct answer – B
Viviparous animals, like most mammals, develop the embryo inside the mother's body and give birth to live offspring.

3. Correct answer – A
Energy flows from plants (producers) to plant-eaters (herbivores) and then to meat-eaters (carnivores).

4. Correct answer – C
Energy flows from plants (producers) to plant-eaters (herbivores) and then to meat-eaters (carnivores).

5. Correct answer – B
Producers (green plants) are the base because they are the only organisms that can convert solar energy into food.

6. Correct answer – B
When one organism eats another, the chemical energy stored in its tissues moves up the food chain.

7. Correct answer – B
Earthworms help break down dead organic matter, returning essential nutrients to the soil.

8. Correct answer – C
Carnivores act as secondary or tertiary consumers by hunting and eating animals lower on the food chain.

9. Correct answer – C
Without producers, there is no energy entering the system, causing herbivores and then carnivores to starve.

10. Correct answer – B
Unlike oviparous animals, the young of viviparous animals do not hatch from an external egg.

COMPETETITIVE CORNER

SINGLE OPTION CORRECT TYPE

1. Correct answer – B
Frogs are vertebrates because they possess an internal skeletal structure featuring a backbone.

2. Correct answer – A
This is the primary anatomical division used to classify the entire animal kingdom.

3. Correct answer – A
Frogs are amphibians because they spend part of their life in water and part on land.

4. Correct answer – B
Gills are the universal adaptation for extracting oxygen from an aquatic environment (though some aquatic mammals use lungs).

5. Correct answer – B
Hollow or pneumatized bones reduce a bird's body weight, making it easier to achieve and maintain flight.

6. Correct answer – A
Carnivores possess specialized teeth, like canines, designed for gripping and ripping animal flesh.

7. Correct answer – A
Migration is a survival strategy to move from resource-poor areas to resource-rich ones during seasonal changes.

8. Correct answer – B
It represents a linear pathway showing who eats whom in an ecosystem.

9. Correct answer – B
Elephants are herbivores that feed exclusively on grasses, bark, fruits, and other plant matter.

10. Correct answer – B
Decomposers like fungi and bacteria act as nature's recyclers, keeping the soil fertile for producers.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct answer – A, C
Humans and Dolphins are mammals that give birth to live young. Frogs and Birds lay eggs (oviparous).

12. Correct answer – A, B, D
Fish have gills/fins, Whales have blubber/flippers, and Ducks have webbed feet and waterproof feathers. Lions are purely terrestrial.

13. Correct answer – A, B, D
Hens (birds), Snakes (reptiles), and Frogs (amphibians) lay eggs. Kangaroos are mammals that give birth to live young (joey).

14. Correct answer – A, B, D
A food chain shows a single linear path of energy flow starting with producers (plants). It doesn't necessarily "end" with large animals, as decomposers follow.

15. Correct answer – A, B, C
Carnivores hunt other animals (A), use sharp teeth/canines for tearing (B), and usually have front-facing eyes for depth perception to track prey (C).

ASSERTION AND REASNING TYPE

16. Correct answer – A
Hollow (pneumatic) bones are a key adaptation that reduces weight without sacrificing structural strength for flight.

17. Correct answer – C
While the body plan description is correct, insects require oxygen and breathe through a system of tubes called spiracles.

18. Correct answer – B
Both are true characteristics of birds, but having feathers is not the *reason* why they lay eggs.

19. Correct answer – A
Food webs are complex because they represent the reality that most organisms belong to multiple food chains simultaneously.

MATCH THE FOLLOWING TYPE

20. Correct answer – (a)-S, (b)-Q, (c)-R, (d)-P

21. Correct answer – (a)-P, (b)-Q, (c)-R, (d)-S

COMPREHENSHION TYPE

Passage1

22. Correct answer - C) To help them fly

23. Correct answer - B) Gills

24. Correct answer - C) Lungs

Passage2

25. Correct answer - B) To reach warmer regions

26. Correct answer - C) Geese

27. Correct answer - C) To find food, warmer climates, or safer places to give birth

Previous year questions

1. Correct answer – A
This is incorrect; the cuttlefish is a soft-bodied mollusk. Animals like porcupines or sea urchins have spines for defense.

2. Correct answer – A
Statement (iii) is incorrect because gills are for breathing, not for increasing swimming speed (fins/tails handle that).

3. Correct answer – C
While X are herbivores and Y are omnivores, all four (sheep, goat, rat, bear) are mammals and therefore viviparous.

4. Correct answer – D
Animals in polar/cold regions (like the polar bear or seal usually shown) rely on "blubber" or fat for insulation.

5. Correct answer – A
(ii) is incorrect as houseflies lay eggs in decaying matter, and (iii) is incorrect as many insects have mobile larval stages (though only adults fly). Both share the 4-stage cycle (Egg → Larva → Pupa → Adult).

3. Food We Eat

Worksheet -1

1. **Correct answer – B**
   Carbohydrates They are the body's preferred and most efficient source of fuel, easily broken down into glucose for energy.
2. **Correct answer – C**
   Proteins provide the essential amino acids required to build muscle, skin, and enzymes while repairing cellular damage.
3. **Correct answer – C**
   It increases stool bulk, regulates bowel movements, modulates gut microbiota, and can help lower blood cholesterol Fiber passes through the digestive system mostly intact, facilitating waste movement and supporting beneficial gut bacteria.
4. **Correct answer – D**
   Vitamin D The body synthesizes Vitamin D through the skin when it is exposed to ultraviolet rays from sunlight.
5. **Correct answer – C**
   Calcium is the primary structural mineral found in the hydroxyapatite crystals that harden bones and teeth.
6. **Correct answer – B**
   Water is the vital medium for all biological fluids, ensuring nutrient transport, temperature regulation, and cellular hydration.
7. **Correct answer – B**
   They serve as a concentrated source of energy, provide thermal insulation, and protect vital organs Fats provide more than double the energy per gram than carbs or protein and act as a protective cushion for the body.
8. **Correct answer – B**
   Fish is a high-quality animal protein source containing all the essential amino acids the body cannot produce on its own.
9. **Correct answer – B/C**
   They act as essential micronutrients regulating biochemical reactions and enhancing immune defense Vitamins function as cofactors for enzymes, enabling critical chemical processes and supporting the body's natural defenses.
10. **Correct answer – C**
    Spinach As a plant-based food, spinach contains cellulose and other fibers that add bulk to the diet, unlike animal products like meat or dairy.

worksheet – 2

1. **Correct answer – B**
   Vitamin A is essential for the production of rhodopsin, a pigment in the retina that allows you to see in low-light conditions.
2. **Correct answer – B**
   e.g., Oranges, Lemons) Citrus fruits are world-famous for their high Vitamin C content, which is vital for immune function and collagen synthesis.
3. **Correct answer – C**
   Rickets This disease causes the softening and weakening of bones in children, typically due to an extreme and prolonged Vitamin D deficiency.

4. Correct answer – C

Fatigue Iron is a key component of hemoglobin, and a lack of it means less oxygen reaches your tissues, leaving you feeling exhausted and weak.

5. Correct answer – C

Without Vitamin D, the body cannot effectively absorb calcium and phosphorus, leading to the bone malformations seen in rickets.

6. Correct answer – C

Iodine The thyroid gland uses iodine to produce hormones; without enough of it, the gland overworks and swells to create a goitre.

7. Correct answer – B

Milk Dairy products like milk are among the most common and bioavailable sources of calcium for maintaining bone density.

8. Correct answer – C

Scurvy results from a lack of Vitamin C, leading to symptoms like bleeding gums, weakness, and poor wound healing.

9. Correct answer – B

Lack of Vitamin B12 Pernicious anemia occurs when the body cannot absorb enough B12, which is necessary for making healthy red blood cells.

10. Correct answer – A

Spinach Dark leafy greens like spinach are excellent plant-based sources of iron, which helps build the hemoglobin needed to prevent anemia.

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worksheet 3

1. Correct answer – B

Tiny living organisms Microbes are microscopic life forms, such as bacteria and fungi, that cannot be seen with the naked eye.

2. Correct answer – C

Bacteria Specific strains of bacteria, like *Lactobacillus*, ferment milk to produce the texture and flavor of yogurt and cheese.

3. Correct answer – B

Release carbon dioxide to make the bread rise and become fluffy As yeast consumes sugar, it releases CO_2 gas bubbles that get trapped in the dough, causing it to expand.

4. Correct answer – B

Fungi Mushrooms are the fleshy, spore-bearing fruiting bodies of a fungus.

5. Correct answer – B

Food poisoning Pathogenic microbes can produce toxins or infections in food that lead to digestive illness.

6. Correct answer – B

Frying is a dry-heat cooking method that uses fats or oils as the heat transfer medium.

7. Correct answer – B

Slows down the growth of harmful microbes Extremely low temperatures inhibit the metabolic activity and reproduction of bacteria that cause spoilage.

8. Correct answer – C

Baking in an oven provides the dry heat necessary for leavening agents (like yeast) to expand and set the bread's structure.

9. Correct answer – B

To draw out moisture and preserve food Salt uses osmosis to pull water out of food cells, creating an environment too dry for most bacteria to survive.

10. Correct answer – B

Drying Dehydration removes the water content that microbes require to grow and remain active.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct answer – B

Carbohydrates These are the body's primary fuel source, breaking down into glucose to power cellular functions.

2. Correct answer – B

Helps in growth and repair Proteins are the building blocks of the body, essential for creating new cells and fixing damaged tissues.

3. Correct answer – A

Carrots are rich in beta-carotene, which the body converts into Vitamin A for vision and skin health.

4. Correct answer – B

A variety of foods from all food groups A balanced diet ensures the body receives the correct proportions of carbs, proteins, fats, vitamins, and minerals.

5. Correct answer – B

Scurvy A lack of ascorbic acid (Vitamin C) prevents collagen production, leading to the bleeding gums and weakness associated with scurvy.

6. Correct answer – A

Boiling involves immersing food in water that has reached its vaporization point (100°C).

7. Correct answer – C

Drying is an ancient preservation technique that eliminates moisture to stop decay.

8. Correct answer – B

Provides energy Carbohydrates are the most readily available source of energy for the brain and muscles.

9. Correct answer – B

Milk and dairy products are highly concentrated sources of calcium, which is vital for maintaining bone density.

10. Correct answer – C

Calcium Vitamin D acts as a key that "unlocks" the intestines, allowing the body to absorb calcium from the food we eat.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct answer – A, B, C

Eggs, fish, and milk are all high-quality protein sources; carrots are primarily a source of vitamins and fiber.

12. Correct answer – A, C

Freezing and canning are preservation methods; frying and boiling are primarily cooking methods (though they can kill microbes temporarily).

13. Correct answer – A, C

Vitamin A is vital for vision (night blindness) and maintaining epithelial tissues (preventing dry skin).

14. Correct answer – A, C

Oranges and tomatoes are citrus/acidic plant sources rich in Vitamin C; carrots are high in Vitamin A, and fish is high in protein/fats.

15. Correct answer – B, C

Steaming and grilling use less water or high direct heat, which helps prevent water-soluble vitamins from leaching out compared to boiling.

ASSERTION AND REASNING TYPE

16. Correct answer – A

A balanced diet requires variety to cover the distinct nutritional roles required for growth and energy.

17. Correct answer – A

Bacteria (like *Streptococcus mutans*) ferment sugars into acids that erode tooth enamel, creating cavities.

18. Correct answer – B

Both are true, but the reason does not explain *why* it preserves; it preserves by creating an acidic or alcoholic environment that inhibits spoilage microbes.

19. Correct answer – A

Microbes like *Lactobacillus* are beneficial because they transform raw milk into shelf-stable, nutritious products like yogurt.

MATCH THE FOLLOWING TYPE

20. Correct answer – (a)—(S), (b)—(Q), (c)—(R), (d)—(P)

21. Correct answer – (a)—(P), (b)—(Q), (c)—(R), (d)—(S)

COMPREHENSHION TYPE

Passage1

22. Correct answer – B

They act as fermenting agents that transform the flavor and physical structure of milk.

23. Correct answer – B

Pathogens produce toxins or infections that lead to illness and food decay.

24. Correct answer – B

Heat (cooking) kills most harmful bacteria, and proper handling prevents cross-contamination.

Passage2

25. Correct answer – A

The passage explicitly lists Carbohydrates, Proteins, Vitamins, and Minerals.

26. Correct answer – B

It ensures the body receives a full spectrum of nutrients rather than just a few.

27. Correct answer – B

Proper nutrition strengthens the immune system and provides the "fuel" for daily activities.

PREVIOUS YEAR QUESTIONS

1. Correct answer – A

Steaming (Assuming the image is of idlis, momos, or dhokla, which are common steaming examples in NSO).

2. Correct answer – A

Sleeping Sickness – It is caused by the protozoan *Trypanosoma*; Dengue is viral, TB is bacterial, and Ringworm is fungal.

3. Correct answer – C

X- Canning, Y- Freezing – Canning is ideal for liquid juices, while freezing keeps peas fresh and green without losing texture.

4. Correct answer – A

2 and 3 only – Stale food is dangerous (not healthy), and we need a *variety* of food (not the same kind) for a balanced diet.

5. Correct answer – D

KIBGAN (BAKING) – Biscuits are made by baking; the unscrambled words are: D) BAKING.

4. Clothes, Safety and First Aid

Worksheet -1

1. **Correct answer – B**
   Cotton is a natural fiber that is breathable and absorbs sweat, which helps keep the body cool during hot summer days.
2. **Correct answer – B**
   Raincoats are made from waterproof materials like plastic or treated synthetic fibers to prevent rainwater from reaching the body.
3. **Correct answer – C**
   Woolen clothes are made from the soft, thick hair (fleece) of animals like sheep, goats, and yaks.
4. **Correct answer – B**
   The Saree is a world-renowned traditional Indian garment consisting of a long drape wrapped around the body.
5. **Correct answer – B**
   Firefighters wear specialized gear made of fibers like Nomex or Kevlar that do not melt or catch fire easily, ensuring their safety in high-heat environments.
6. **Correct answer – B**
   The process starts with spinning raw fibers into yarn/thread, which is then woven into fabric and finally stitched into clothes.
7. **Correct answer – B**
   Due to the nature of their work involving fire and extreme heat, these professionals require specialized flame-resistant uniforms.
8. **Correct answer – B**
   Nylon is a synthetic fiber that does not absorb water easily, making it an ideal choice for raincoats and umbrellas.
9. **Correct answer – C**
   Wool is a poor conductor of heat and traps air between its fibers, which prevents body heat from escaping into the cold winter air.
10. **Correct answer – B**
    Unlike cotton, wool, and silk (which come from plants or animals), polyester is a man-made synthetic fiber derived from chemical compounds.

worksheet – 2

1. **Correct answer – C**
   Wool is a natural fiber harvested from animals, whereas Nylon, Polyester, and Acrylic are man-made synthetic fibers.
2. **Correct answer – C**
   Cotton fibers grow as "bolls" around the seeds of the cotton plant, which are then harvested and processed.
3. **Correct answer – B**
   Silk is a natural protein fiber produced by silkworms, prized for its shimmering appearance and smooth texture.
4. **Correct answer – C**
   Synthetic fibers like polyester are engineered to maintain their shape and require little to no ironing compared to natural fibers.

5. **Correct answer – C**
   Spinning creates the thread (yarn) by twisting raw fibers together, and weaving uses those threads to construct the actual fabric.
6. **Correct answer – B**
   Woolen fibers are delicate and can shrink or lose their shape if exposed to harsh chemicals or high heat.
7. **Correct answer – B**
   Mothballs (naphthalene) or dried neem leaves are used to repel insects like silverfish and moths that feed on natural fibers.
8. **Correct answer – B**
   The correct order is: Collecting raw material, spinning it into yarn, weaving the yarn into fabric, and finally stitching it into a garment.
9. **Correct answer – B**
   Silk is sensitive to UV rays, which can break down the protein fibers and cause the colors to lose their luster.
10. **Correct answer – B**
    Most synthetic fibers are plastic-based and do not decompose naturally, leading to environmental accumulation and pollution.

worksheet 3

1. **Correct answer – B**
   A first aid box is a collection of supplies and equipment used to give immediate medical treatment before professional help arrives.
2. **Correct answer – B**
   Cleaning the wound is the first step to remove dirt and bacteria to prevent infection.
3. **Correct answer – A**
   Cotton balls are used to apply antiseptic liquids or gently dab away blood and dirt from a wound.
4. **Correct answer – B**
   Leaning forward prevents blood from going down the throat, and pinching the soft part of the nose helps stop the bleeding.
5. **Correct answer – C**
   This position helps blood flow back to the brain, which can help the person regain consciousness.
6. **Correct answer – B**
   Beginners lack the strength and skill to manage deep water safely; jumping in without experience is a major drowning risk.
7. **Correct answer – B**
   Synthetic fibers are chemically engineered to last long and maintain their shape better than most natural fibers.
8. **Correct answer – A**
   In countries where vehicles drive on the left (like India), looking right first checks the nearest lane of traffic, followed by left, then right again for final safety.
9. **Correct answer – B**
   The immediate priority is to prevent drowning by removing them from the water and ensuring their airway is clear.
10. **Correct answer – C**
    Helmets act as a shock absorber for the skull, significantly reducing the risk of serious brain injury during a fall.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer – B**
   Cotton allows air circulation and absorbs moisture, making it ideal for hot weather.
2. **Correct answer – C**
   Firefighters use specialized "turnout gear" made of flame-resistant synthetic fibers to survive extreme temperatures.
3. **Correct answer – B**
   These are traditional Indian garments that reflect the cultural heritage of the country.
4. **Correct answer – B**
   Wool is a natural fiber obtained from the fleece of animals like sheep, goats, or camels.
5. **Correct answer – B**
   Proper storage prevents moisture damage and fading; silk should also be kept away from direct sunlight.
6. **Correct answer – B**
   Synthetic materials like plastic or nylon do not allow water to pass through, keeping the wearer dry.
7. **Correct answer – B**
   Safety rules are designed to organize the movement of people and vehicles to prevent collisions and injuries.
8. **Correct answer – C**
   Small fires can often be contained immediately if the right tools (like an extinguisher) are used safely and quickly.
9. **Correct answer – B**
   Bandages are essential for covering wounds, stopping bleeding, and supporting injured joints.
10. **Correct answer – B**
    The "RICE" method (Rest, Ice, Compression, Elevation) is the standard first aid for sprains to reduce swelling and pain.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct answer – A, C**
    Nylon and Rayon are synthetic (man-made). Wool and Silk are natural animal fibers.
12. **Correct answer – A, B, D**
    Never store wet clothes in a closed space (C) as it leads to mold and bad odors.
13. **Correct answer – A, B, C**
    Bandages, antiseptic liquid, and cotton balls are medical essentials. Snacks are not a first aid item.
14. **Correct answer – A, C**
    Safety rules in swimming are primarily designed to prevent life-threatening accidents like drowning.
15. **Correct answer – A, C**
    Synthetic fibers melt easily, so high heat (A) must be avoided. Mild detergents (C) preserve the fiber quality.

ASSERTION AND REASNING TYPE

16. **Correct answer – A**
    Cotton's porous nature allows it to absorb sweat and evaporate it, cooling the skin.

17. **Correct answer – A**
    Helmets provide a protective layer that absorbs the impact energy during a crash.
18. **Correct answer – A**
    Wool fibers have natural crimps that trap air; air is a poor conductor of heat, keeping body warmth in.
19. **Correct answer – C**
    First aid should never be delayed; its purpose is to provide "immediate" help.
20. **Correct answer – C**
    Synthetic fibers are non-biodegradable because they are made from petroleum-based chemicals, not natural raw materials.

MATCH THE FOLLOWING TYPE

21. **Correct answer –** (a)-Q, (b)-P, (c)-R, (d)-S
22. **Correct answer –** (a)-R, (b)-Q, (c)-P, (d)-S

COMPREHENSHION TYPE

Passage1

23. **Correct answer – B**
    They absorb sweat and allow air circulation
24. **Correct answer – C**
    It absorbs sweat
25. **Correct answer – C**
    Woolen clothes

Passage2

26. **Correct answer – B**
    Immediate help given to an injured person
27. **Correct answer – B**
    Antiseptic
28. **Correct answer – B**
    Bandage

Previous year questions

1. **Correct answer – A**
   The Nauvari (nine-yard) saree and the Dhoti-Kurta are iconic traditional garments of Maharashtra.
2. **Correct answer – D**
   In option D, Wool, Jute, and Linen are all natural fibers, while Lycra is synthetic. Circling "Wool" as the odd one out is incorrect because it belongs to the natural group.
3. **Correct answer – C**
   Haryana, Rajasthan, Andhra Pradesh, Himachal Pradesh
4. **Correct answer – A**
   Nagaland- The distinctive colorful woven patterns and traditional shawls shown in the image are characteristic of the Naga tribes.
5. **Correct answer – D**

5. Human Body

Worksheet -1

1. **Correct answer – B**
   Mouth Digestion officially begins here as food is physically broken down by teeth and chemically altered by saliva.
2. **Correct answer – A**
   Stomach The stomach produces hydrochloric acid and gastric juices (enzymes) that break down proteins into a semi-liquid state called chyme.
3. **Correct answer – B**
   To break food into smaller pieces Teeth, perform mechanical digestion, increasing the surface area of food for enzymes to work on later.
4. **Correct answer – C**
   Incisors These are the sharp, chisel-shaped teeth located at the front of the mouth, specifically designed for cutting and biting into food.
5. **Correct answer – C**
   Small intestine While digestion starts elsewhere, the small intestine is where most chemical breakdown finishes and where nutrients pass into the bloodstream.
6. **Correct answer – A**
   To help break down food chemically Saliva contains the enzyme amylase, which begins the breakdown of starches into simpler sugars.
7. **Correct answer – B**
   Stomach The stomach acts as a temporary reservoir, holding and mixing food with acids for several hours.
8. **Correct answer – C**
   III → I → II The journey starts with chewing (III), followed by swallowing through the food pipe (I), and ends with waste processing in the large intestine (II).
9. **Correct answer – B**
   Large intestine Its main job is to reclaim water from undigested material, compacting the remaining waste into solid stool.
10. **Correct answer – D**
    Least likely to tear: Canines; Most likely to grind: Molars Note: Technically, canines *are* used to tear, but in the context of this specific question's logic regarding grinding, **Molars** are the definitive grinders used at the end of the process.

worksheet – 2

1. **Correct answer – B**
   To deliver oxygen to the body and remove carbon dioxide The system facilitates gas exchange between the air and the blood.
2. **Correct answer – B**
   Mouth or Nose These are the external openings that allow air to enter the respiratory tract.
3. **Correct answer – C**
   Alveoli These tiny air sacs are surrounded by capillaries where oxygen diffuses into the blood and carbon dioxide leaves it.
4. **Correct answer – B**
   Diaphragm This dome-shaped muscle contracts to create a vacuum that pulls air into the lungs.

5. **Correct answer – B**
   Carbon dioxide This is the waste product of cellular respiration that is expelled from the lungs.
6. **Correct answer – A**
   To transport oxygen, nutrients, and waste products The, system acts as the body's internal delivery and disposal network.
7. **Correct answer – C**
   Heart The heart is a muscular pump that generates the pressure needed to keep blood moving.
8. **Correct answer – C**
   II → IV → I → III Air enters the **Nose/Mouth**, travels down the **Trachea**, branches into the **Bronchi**, and ends in the **Alveoli**.
9. **Correct answer – C**
   To exchange oxygen and nutrients with cells Capillaries are microscopic vessels with thin walls that allow substances to pass between blood and tissues.
10. **Correct answer – C**
    Platelets are cell fragments that clump together to form a plug (clot) at the site of a wound.

COMPETITIVE CORNER

SINGLE OPTION CORRECT TYPE

1. **Correct answer – B**
   To grind and break food into smaller pieces Teeth perform mechanical digestion, preparing food for chemical breakdown.
2. **Correct answer – C**
   Small intestine While it starts in the mouth and stomach, the majority of chemical digestion is completed in the small intestine.
3. **Correct answer – B**
   Small intestine Its lining is covered in villi, which provide a massive surface area for nutrient absorption.
4. **Correct answer – B**
   Amylase Salivary amylase is the enzyme that specifically begins breaking down complex starches into simple sugars.
5. **Correct answer – C**
   Alveoli These are the functional units of the lungs where the actual gas exchange occurs.
6. **Correct answer – B**
   To deliver oxygen to the blood and remove carbon dioxide This defines the primary physiological purpose of breathing.
7. **Correct answer – B**
   Diaphragm By moving up and down, it changes the pressure in the chest cavity to move air.
8. **Correct answer – B**
   Arteries carry blood **away** from the heart; most carry oxygenated blood (except the pulmonary artery).
9. **Correct answer – B**
   Heart, It functions as a double pump to move blood to both the lungs and the rest of the body.

10. **Correct answer – B**
    To carry oxygen red blood cells, contain hemoglobin, a protein that binds to oxygen for transport to the body's cells.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct answer – A, B, C**
    The kidneys, ureters, and bladder are the core components of the urinary (excretory) system. (Note: While Lungs excrete CO_2, they are primarily categorized under the Respiratory system).
12. **Correct answer – A, B, D**
    The digestive system breaks down food, absorbs nutrients, and eliminates solid waste (defecation).
13. **Correct answer – A, B, D**
    The Lungs, Trachea (windpipe), and Nose are all primary structures for breathing and gas exchange.
14. **Correct answer – A, B, C**
    RBCs transport oxygen, WBCs act as the immune defense, and Platelets stop bleeding. Plasma is the liquid medium, not a digestive agent.
15. **Correct answer – A, B, C, D**
    All these systems are interdependent; for example, the respiratory system provides oxygen that the circulatory system delivers to the digestive system to function.

ASSERTION AND REASNING TYPE

16. **Correct answer – A**
    Villi create a large surface area, which is the specific structural reason why the small intestine is so efficient at absorbing nutrients.
17. **Correct answer – D**
    The Assertion is false because the heart pumps **oxygen-poor** (deoxygenated) blood to the lungs to get oxygen. The Reason correctly describes this process.
18. **Correct answer – A**
    Haemoglobin is the specific iron-rich protein that allows RBCs to pick up and carry oxygen molecules.
19. **Correct answer – A**
    Respiration isn't just about taking oxygen in; it is equally about the removal of metabolic waste (CO_2).
20. **Correct answer – B**
    Both statements are true, but the fact that kidneys regulate water balance is a separate function and not the explanation for how they filter toxins.

MATCH THE FOLLOWING TYPE

21. **Correct answer –** (a)—(R), (b)—(S), (c)—(P), (d)—(Q)
22. **Correct answer –** (a)—(S), (b)—(P), (c)—(Q), (d)—(R)
23. **Correct answer -** (a)—(Q), (b)—(R), (c)—(S), (d)—(P)
24. **Correct answer -** (a)—(Q), (b)—(P), (c)—(S), (d)—(R)

COMPREHENSHION TYPE

Passage1

25. **Correct answer – B**
    Oxygen and nutrients – As stated in the passage, blood acts as the transport medium for these essentials.

26. **Correct answer – C**
    Respiratory system – This system brings in air so oxygen can diffuse into the bloodstream.
27. **Correct answer – C**
    To break down food and absorb nutrients – This is the primary biological role of the digestive tract.

Passage2

28. **Correct answer – B**
    To filter waste from the blood – Kidneys act as the body's natural filtration system.
29. **Correct answer – C**
    Toxins and waste – They clean the blood to prevent the buildup of harmful substances.
30. **Correct answer – C**
    Urine – This is the final liquid byproduct of the kidney's filtration process.

Previous year questions

1. **Correct answer – D**
   Kidneys – They purify the blood by removing metabolic wastes (often shown as bean-shaped organs in diagrams).
2. **Correct answer – D**
   Excretion – (Assuming the pictures show organs like the kidneys and skin, or kidneys and lungs, which both remove waste).
3. **Correct answer – D**
   Z – In standard digestive system diagrams, **Z** usually points to the **Anus**, where faeces are expelled.
4. **Correct answer – C**
   Secretion of an acid will not take place – If **X** labels the **Stomach**, its removal would stop the production of hydrochloric acid (HCl).
5. **Correct answer – B**
   Canines – These sharp, pointed teeth are specifically designed for tearing food (like meat).

6. Matter around us

Worksheet-1

1. **C) Melting**
   Melting is the process where a solid (like ice) turns into a liquid when heated.
2. **D) Solid**
   Solids have a fixed shape and fixed volume because their particles are tightly packed.
3. **B) Condensation**
   Condensation happens when gas cools and changes into a liquid (like water droplets forming from steam).
4. **C) Vaporisation**
   Vaporisation is when a liquid changes into a gas due to heating.
5. **B) They move slower and get closer together**
   When water freezes, the molecules lose energy, slow down, and form a solid structure.
6. **C) Sublimation**
   Sublimation is when a solid changes directly into gas without becoming liquid (e.g., dry ice).
7. **B) Liquid**
   Liquids take the shape of their container but have a fixed volume.
8. **A) There is less air pressure, so the liquid needs less heat to boil**
   Lower air pressure makes it easier for liquid molecules to escape as gas.
9. **C) Deposition**
   Deposition is when gas changes directly into solid (water vapor → ice).
10. **C) Liquid → Gas: Evaporation**
    Evaporation is the correct term for liquid changing into gas. The other options are mismatched.

Worksheet-2

1. **B) Volume**
   Volume is the amount of space an object takes up.
2. **C) Solubility**
   Solubility is the ability of a substance to dissolve in a liquid.
3. **B) Thermal expansion**
   Thermal expansion is when materials expand (increase in size) on heating.
4. **A) Copper – Rubber**
   Copper conducts electricity, while rubber does not (insulator).
5. **C) Copper**
   Copper is a very good conductor of electricity and is flexible.
6. **B) Opacity**
   Opaque materials do not allow light to pass through.
7. **C) Melting point**
   Melting point is the temperature at which a solid turns into a liquid
8. **A) Elasticity**
   Elasticity is the ability of a material to regain its shape after stretching.
9. **B) Ductility**
   Ductility is the ability to be stretched into thin wires.

10. C) Lustre
Lustre means the shiny appearance of metals.

Worksheet-3

1. B) Solvent
The solvent is the substance that dissolves another substance (sugar).

2. C) Heterogeneous mixture
In a heterogeneous mixture, components are visible (like sand and water).

3. C) Sand settles at the bottom
Sand is insoluble, so it settles down instead of dissolving.

4. A) Salt water
Salt water contains salt + water physically combined.

5. A) Saltwater
Homogeneous mixtures look uniform throughout.

6. B) A chemical reaction
In mixtures, substances combine physically, not chemically.

7. C) Air
Air is a mixture of gases like oxygen, nitrogen, etc.

8. B) Solute: Sugar and Lemon Juice; Solvent: Water
Water dissolves sugar and lemon juice → so water is the solvent.

9. C) Mix completely and look like one substance
Solutions are homogeneous and uniform.

10. C) Solute
The solute is the substance that gets dissolved.

Worksheet-4

1. A) Filtration
Sand is insoluble in water, so it can be separated using a filter.

2. B) Magnetic Separation
Iron is magnetic, while sulphur is not, so a magnet can separate them.

3. B) Difference in size; Sieve
Poppy seeds are larger than flour particles, so sieving works best.

4. B) Sieving
Different particle sizes allow separation using a sieve.

5. D) Letting them settle
Oil and water are immiscible and form layers; oil floats on water and can be separated.

6. D) Sieving
Rice grains are much larger than salt particles.

7. B) Magnetic Separation
Neither sand nor salt is magnetic, so this method won't work.

8. B) Because of its density
Chaff is lighter (less dense) and is blown away by wind, while grains fall down.

9. B) Physical change
Physical changes (like melting/freezing) can usually be reversed.

10. C) Centimeter
Centimeter is a unit of length; others measure mass, volume, or time.

Competitive Corner

1. D) It is mixed in any proportion by mass

Compounds are formed by elements combining in a fixed ratio by mass, not any proportion.

They are pure substances and cannot be separated by physical methods.

2. B) iodine is the solute and alcohol is the solvent

A solute is the substance that gets dissolved.

A solvent is the substance that does the dissolving.

In tincture of iodine, iodine dissolves in alcohol, so:

Iodine = solute

Alcohol = solvent

3. C) (iii) only

(iii) Saline Solution → actually homogeneous, not heterogeneous

4. C) Mercury, bromine

Mercury → liquid metal

Bromine → liquid non-metal

5. A) Centrifugation

Centrifugation spins the mixture to separate lighter cream from heavier milk.

6. B) Metal

Metals are extracted and processed by humans → considered man-made in this context.

7. B) Distilled water

Distilled water contains only water (no impurities).

Air, steel, brass → mixtures.

8. C) Both a and b *(most appropriate)*

Water can exist as liquid and vapor depending on temperature.

9. C) Gas

Gas particles have very weak attraction and move freely/randomly.

10. B) Naphthalene

Naphthalene changes directly from solid to gas (used in mothballs).

11. B) Measuring Jug/Beaker

These are designed to measure liquid volumes accurately.

One or more than one option:

12. A) Sieving

B) Handpicking

- Sieving separates particles of different sizes using a mesh.
- Handpicking works when components are large enough to pick by hand.
- Filtration is based on particle size *in liquids*, but mainly for solid-liquid separation.
- Distillation is based on boiling point, not size.

13. A) Separating funnel

- A separating funnel allows liquids of different densities to form layers and be separated.
- Distillation is used for miscible liquids, not immiscible ones.

14. A) Filtration

B) Sedimentation

- Sedimentation: Sand settles at the bottom.
- Filtration: Separates sand using a filter.
- Winnowing and sieving are not suitable for this mixture.

15. A) Salt solution
B) Air
D) Sugar dissolved in water

- Homogeneous mixtures are uniform throughout.
- Oil and water is heterogeneous (forms layers).

16. A) The composition is not uniform
B) Different components can often be seen
D) They may have two or more visible parts

- Heterogeneous mixtures are non-uniform and have visible components.
- Option C is incorrect because they do not form a single phase.

Assertion and reason:

17. Answer: (A)
Explanation:

- Assertion is true: Condensation is the process where a gas turns into a liquid (e.g., water vapor → water).
- Reason is true: Gas particles lose energy, slow down, and come closer, forming a liquid.

18. Answer (C)
Explanation:

- Assertion is false: When ice melts, it changes from solid to liquid, *not gas*.
- Reason is true: Melting happens when a substance absorbs heat.

Match the following:

19. Explanation
Length: Measured in meters (m) in the SI system; it represents the distance between two points.
Mass: Measured in kilograms (kg); it represents the amount of matter in an object.
Time: Measured in seconds (s); it is the progression of events from the past to the future.
Temperature: While Kelvin is the SI base unit, Degree Celsius (°C) is a standard metric unit used globally to measure how hot or cold an object is.

20. Explanation

- Glass window (Transparency): Glass allows light to pass through clearly, making it transparent.
- Wood (Opacity): Wood is an opaque material; light cannot pass through it at all.
- Silver (Lustre): Lustre refers to the way light interacts with the surface of a crystal, rock, or metal. Silver is known for its high metallic shine or "lustre."
- Red apple (Colour): This matches the visual perception of the object. The "red" specifically identifies the characteristic colour of the fruit's skin.

Comprehension:

Passage-1

21. A) Solid
Solids have tightly packed particles, so they keep both their shape and volume.

22. C) Gas
Gas particles move freely and spread out to fill any container completely.

23. B) Liquid
Liquids can flow and take the shape of the container, but their volume remains fixed.

Passage-2

24. A) Homogeneous mixture

Homogeneous mixtures look the same throughout (uniform composition).

25. B) Heterogeneous mixture

In heterogeneous mixtures, components are visible and not evenly mixed.

26. A) Homogeneous mixture

Salt dissolves completely in water, forming a uniform solution**.**

Previous Year Questions

1. Answer: D. d

To solve this, we must identify the state of matter of honey. Honey is a liquid.

- Definite Weight: All matter (solids, liquids, and gases) has weight.
- Definite Shape: Only solids have a definite shape. Liquids take the shape of their container.
- Middle Circle (Left): This circle likely represents "Definite Volume" (though the text is partially cut off, this is the standard third property).
- The Breakdown: * 'b' is the intersection of all three (Solid).
  - 'a' and 'c' represent intersections of only two properties.
  - 'd' represents an object that has Definite Weight and Definite Volume but NO Definite Shape. This perfectly describes a liquid like honey

2. Answer: C. 2, 4 and 5 only

Processes that "consume heat" are endothermic. This means they need energy (heat) to be added to change the state of matter.

1. Condensation: Releases heat (Gas →Liquid).
2. Evaporation: Consumes heat (Liquid →Gas).
3. Freezing: Releases heat (Liquid →Solid).
4. Melting: Consumes heat (Solid →Liquid).
5. Boiling: Consumes heat (Liquid →Gas).

3. Answer: D. Volume of shampoo

Explanation: The object labeled X is a graduated cylinder (or measuring cylinder).

- Function: A graduated cylinder is a standard laboratory tool used specifically to measure the volume of liquids.
- Why the others are wrong: * Temperature is measured with a thermometer.
  - Weight is measured with a weighing scale or spring balance.
  - Length is measured with a ruler or measuring tape.
- Since shampoo is a liquid, its volume is correctly measured using this tool

4. Correct Answer: B) Glass

- A material that allows light to pass is transparent.
- Having fixed shape and volume means it is a solid.
- Glass is a transparent solid.
- Marble is solid but not transparent, water has no fixed shape, and carbon dioxide is a gas.

5. Step 1: Strike out every alternate letter (starting from first letter):

Letters: P M O E S T Q A Z L

Remove: P, O, S, Q, Z

Remaining: M E T A L

A) Crayons

- Metals are used in televisions, pressure cookers, bridges.
- Crayons are made from wax, not metal.

7. Force, Work, Energy and Machines

Worksheet-1

1. C) To make work easier
Simple machines reduce effort by changing the direction or magnitude of force, making work easier.

2. B) Lever
A lever is a rigid bar that rotates around a fixed point called a fulcrum.

3. C) Ramp
A ramp is an inclined plane that helps move objects to a higher or lower level with less effort.

4. C) Wedge
A wedge is used to cut or split objects by applying force to separate materials.

5. C) Screw
A jar lid works as a screw because a screw is a simple machine that holds things together (like a bottle cap) or lifts objects when turned

6. C) Wooden block
Magnetic force acts only on magnetic materials like iron and steel, not on wood.

7. (A)
Picture A shows magnets holding papers against a metal refrigerator door, which is a direct demonstration of magnetic attraction.

8: B) Because of gravity
Gravity pulls objects toward the Earth, preventing us from floating like astronauts in space.

9. B) Muscular force
Muscular force is applied by our body muscles to lift or carry objects.

10. B) No friction
Oily surfaces reduce friction, causing slipping because there is less grip between the foot and the floor.

Worksheet-2

1. B) Energy
Work is done when energy is used to apply force and cause movement. Energy is the ability to do work.

2. C) Kicking a football
Work is done when force causes displacement. Kicking a football applies force and moves it.

3. C) Water Energy
Standard forms include heat, light, and mechanical energy. “Water energy” is not a standard form (it is usually called hydro energy).

4. B) Light Energy
Light energy enables us to see objects as it reflects from them into our eyes.

5. B) Mechanical Energy
A moving car has kinetic energy, which is a form of mechanical energy.

6. C) Heat Energy
Heat energy increases the temperature of water until it reaches boiling point.

7. B) A fan working
A fan uses electrical energy to rotate its blades.

8. (B)
Sun produces both light and heat energy.

9. A) Objects vibrate
Sound energy is produced when objects vibrate and create sound waves.

10. B) Sound Energy
A microphone uses sound energy as input and converts it into electrical signals.

Worksheet -3

1. C) Sunlight
Renewable energy sources are naturally replenished. Sunlight is continuously available and does not get exhausted.

2. D) Coal
Coal is a fossil fuel that takes millions of years to form and will eventually run out.

3. C) Petroleum
Petroleum is a fossil fuel and cannot be replaced quickly, so it is non-renewable.

4. C) They cause pollution
Non-renewable sources like coal and petroleum release harmful gases, causing pollution and environmental damage.

5. B) Wind Energy
Wind energy is produced by the movement of air and is used to generate electricity using windmills.

6: A) Biomass Energy
Biomass energy comes from burning organic materials like dead plants and animal waste.

7. B) Renewable
Wind energy is naturally replenished and does not get exhausted, so it is renewable.

8. B) Water
Hydroelectric power plants use flowing or falling water to generate electricity.

9. C) Solar Energy
Solar energy is clean, renewable, and does not produce pollution, making it best for the environment.

10. C) Renewable energy is eco-friendly
Renewable sources like solar and wind cause less pollution and are environmentally friendly.

Competitive Corner

Single Correct Option Type

1. A) Object should move
Work is done when force causes displacement (movement). If no movement, no work is done.

2. D) Weight energy
Heat, sound, and light are forms of energy. Weight is a force, not a form of energy.

3. C) Heat energy
Rubbing hands creates friction, which produces heat energy.

4. A)
A knife is an example of a wedge, which is a simple machine that tapers to a thin edge and is used for splitting or cutting.

5. B) Sound energy
Sound energy is responsible for producing and hearing sounds.

6. B) Natural gas
Cooking gas (LPG/CNG) comes from natural gas, a non-renewable fossil fuel.

7. B) They cause pollution
Burning fossil fuels releases harmful gases causing air pollution.

8. B) Bulb
A bulb converts electrical energy into light energy.

9. B) Potential energy
A stretched rubber band stores elastic potential energy due to its position/shape.

10. B) Electrical energy changes to light energy
In a torch, battery supplies electrical energy which is converted into light energy.

ONE OR MORE THAN CORRECT OPTION TYPE

11. A, B, C
Sound, heat, and electrical energy are forms of energy. Gravity is a force, not energy.

12. A, B
Wind and solar energy are renewable. Petroleum and coal are non-renewable.

13. A, B, C
Guitar, drums, and running water produce sound. A light bulb mainly produces light.

14. A, B, D
Television, washing machine, and mobile phone use electrical energy. Candle uses chemical energy.

Assertion and Reason Type

15. C) Assertion is true and Reason is false.
Windmills convert wind energy to electricity (true), but wind is renewable, not non-renewable (false).

16. A) Both Assertion and Reason are true and Reason is a correct explanation of Assertion.
Vibrations produce sound, and these vibrations travel as waves to our ears.

Matching Type Questions

17. A–S, B–Q, C–R, D–P
Explanation: Solar → panels, Wind → windmills, Biomass → biogas cooking, Petroleum → vehicles.

18. A–Q, B–P, C–R, D–S
Explanation: Lever → scissors, Wedge → knife, Pulley → flagpole, Wheel & axle → bicycle.

Comprehension Type Questions

19. A) Coal, water, wind, and solar energy
These are common sources used to generate electricity.

20. B) It powers lights, fans, refrigerators, and appliances
Electricity is essential for operating daily-use devices.

21. B) Wind and solar energy
These sources are clean and renewable, causing minimal pollution.

22. B) Pulley
A pulley is used to lift objects by changing the direction of force.

23. C) It changes the direction of force and makes lifting easier
Pulling down lifts the load up, reducing effort.

24. C) No, because lifting heavy bricks directly is difficult
Without a pulley, more force is required, making lifting harder.

Previous Year Questions

1. A) Inserting and removing a plug into a socket
Inserting a plug needs a push, and removing it needs a pull. So this action shows both pushing and pulling forces together.

2. B) A screw is a simple machine that has a grooved wheel and rope running in the groove of the wheel.
This statement is incorrect because it describes a pulley, not a screw. A screw is an inclined plane wrapped around a cylinder.

3. B) Steering wheel
A wheel and axle has a large wheel attached to a smaller axle and both rotate together

4. (C)
(i) “A beam or rod arranged in such a manner that it can move freely around a fixed point” describes a lever.
(ii) “Made up of two inclined planes that meet to form a sharp edge” describes a wedge.
A) Ramp is an inclined plane, bottle cap is screw.
B) Screwdriver is wheel and axle, claw hammer is lever.
C) Scissor is mainly a lever, fork can act like a wedge.
D) Bicycle pedal is not the intended lever example, bottle opener is lever.

5. A) Energy that comes from flowing water is called hydro energy.
Hydro energy is energy obtained from moving or flowing water. The other options are incorrect because nutcracker is a lever, bottle opener is a lever, and geothermal energy comes from heat inside the Earth.

8. Air and Water

Worksheet – 1

1. **Correct Answer: B**
   The atmosphere is the layer of air surrounding the Earth. It contains gases like nitrogen and oxygen that support life.
2. **Correct Answer: A**
   The mesosphere is the coldest layer of the atmosphere. Temperatures here can drop to very low levels, making it the coldest region.
3. **Correct Answer: A**
   The ozone layer blocks harmful ultraviolet (UV) radiation from the Sun. Without it, more UV rays would reach Earth and harm living organisms.
4. **Correct Answer: B**
   The atmosphere protects Earth by absorbing and scattering harmful solar radiation. It acts like a protective shield.
5. **Correct Answer: A**
   The atmosphere is responsible for weather phenomena like rain, wind, and storms. These occur mainly in the lower layers.
6. **Correct Answer: B**
   Although temperatures are high in the thermosphere, the air is very thin. This means there are fewer particles to transfer heat, so it doesn't feel hot.
7. **Correct Answer: B**
   Most shooting stars (meteors) burn up in the mesosphere due to friction with air particles. This prevents them from reaching Earth's surface.
8. **Correct Answer: B**
   Soil erosion is the removal of the fertile top layer of soil by wind or water. It reduces soil fertility and affects agriculture.
9. **Correct Answer: B**
   The atmosphere helps keep Earth warm by trapping heat from the Sun. This process is known as the greenhouse effect.
10. **Correct Answer: A**
    The correct order from lowest to highest is: Troposphere → Stratosphere → Mesosphere → Thermosphere.

Worksheet – 2

1. **Correct Answer: B**
   Air is not a single gas but a mixture of gases like nitrogen, oxygen, carbon dioxide, and others. It is invisible and surrounds the Earth.
2. **Correct Answer: C**
   Nitrogen is the most abundant gas in the air, making up about 78% of the atmosphere. Oxygen is the second most abundant.
3. **Correct Answer: B**
   Statement (i) is correct because nitrogen is more than oxygen. Statement (ii) is also correct because oxygen is much more than carbon dioxide, but (iii) is incorrect.
4. **Correct Answer: B**
   Plants use carbon dioxide during photosynthesis to make food. This process also releases oxygen into the air.

5. **Correct Answer: B**
The balloon filled with air adds weight, showing that air has mass and therefore weight. Even though air is invisible, it still occupies space and has weight.
6. **Correct Answer: C**
Water vapour is present in small amounts in air and plays a key role in cloud formation and rainfall. It changes into liquid droplets to form clouds.
7. **Correct Answer: C**
The balloon does not expand because the bottle is already filled with air. This shows that air occupies space.
8. **Correct Answer: C**
Moving air is called wind. It can vary from a gentle breeze to a strong storm.
9. **Correct Answer: B**
Air pressure is the force exerted by air on all objects. It acts in all directions around us.
10. **Correct Answer: B**
Burning fuels in vehicles and factories releases harmful gases into the air. These pollutants cause air pollution.

Worksheet – 3

1. **Correct Answer: B**
Water pollution is when harmful substances like chemicals, waste, or sewage make water dirty and unsafe. This affects aquatic life and human health.
2. **Correct Answer: B**
Dumping garbage in rivers introduces harmful materials into water bodies. This contaminates the water and causes pollution.
3. **Correct Answer: A**
Oil spills in seas and oceans directly contaminate water bodies, making them a cause of water pollution. The other options (smoke, dust, harmful gases) are causes of air pollution, not water pollution.
4. **Correct Answer: C**
Throwing garbage in bins prevents waste from entering water bodies. This helps reduce water pollution.
5. **Correct Answer: B**
Boiling water kills harmful bacteria and makes it safe for drinking. High temperature destroys microorganisms.
6. **Correct Answer: C**
Chlorination is used to kill harmful bacteria and germs in water. It disinfects water and makes it safe to use.
7. **Correct Answer: A**
Statements 1 and 2 are correct: pure water has no colour, taste, or smell, and it exists in three states. Statement 3 is incorrect because water can dissolve substances like salt and sugar.
8. **Correct Answer: A**
Evaporation is the process where water changes from liquid to vapor due to heat. It is part of the water cycle.
9. **Correct Answer: B**
Factory waste introduces harmful substances into water. These pollutants harm aquatic life and can lead to fish deaths.

10. Correct Answer: B
Oceans are the largest source of water on Earth, containing about 97% of all water.

Competitive Corner

Single Option Correct Type

1. Correct Answer: D
"Firesphere" is not a real layer of the atmosphere. The actual layers include troposphere, stratosphere, mesosphere, thermosphere, and exosphere.

2. Correct Answer: C
Loamy soil contains a balanced mixture of sand, clay, humus, air, and water. It is ideal for plant growth.

3. Correct Answer: C
Humans live in the troposphere, the lowest layer of the atmosphere. This is where weather changes occur.

4. Correct Answer: B
Air pollution happens when harmful gases, dust, and smoke mix with clean air. This makes the air unsafe to breathe.

5. Correct Answer: A
Evaporation connects air and water as water changes into vapor and enters the atmosphere. It is part of the water cycle.

6. Correct Answer: B
Pollution traps heat in the atmosphere, leading to global warming. This increases Earth's average temperature.

7. Correct Answer: B
Using public transport reduces the number of vehicles on the road. This lowers air pollution.

8. Correct Answer: B
Weather occurs mainly in the troposphere. If the airplane is above weather, it is flying in the stratosphere.

9. Correct Answer: B
Condensation is when water vapor cools and turns into tiny droplets, forming clouds. It is a key step in the water cycle.

10. Correct Answer: C
Filtration is a method of water purification, not a natural step in the water cycle. The cycle includes evaporation, condensation, and precipitation.

11. Correct Answer: B
Sunlight in solar purification kills bacteria and algae. This helps make water safer to drink.

12. Correct Answer: D
Evaporation, condensation, and precipitation are parts of the water cycle. Filtration is not, so it is the odd one out.

One or More than One Correct Option Type:

13. Correct Answer: A, C, D
Oil spills, sewage discharge, and factory waste contaminate water and cause pollution. Using too much water leads to wastage, not pollution.

14. Correct Answer: A, C, D
Air has mass, is invisible, and can move (wind). The statement that air does not take up space is incorrect—air does occupy space.

15. Correct Answer: A, B, D
Boiling, filtering, and distillation are common methods to purify water. Adding salt does not purify water.

16. Correct Answer: A, B, C, D
Air contains oxygen, carbon dioxide, nitrogen, and small amounts of other gases like hydrogen. These gases together form the atmosphere.

17. Correct Answer: A, B, C
Air pollution causes breathing problems, harms plants, and contributes to climate change. It does not produce clean air.

Assertion and Reason Type

18. Correct Answer: A
The assertion is true because air is a mixture of different gases. The reason correctly explains this by naming gases like oxygen, nitrogen, and carbon dioxide that make up air.

19. Correct Answer: A
The assertion is true as evaporation is the change of water from liquid to gas. The reason is also true and correctly explains that heat from the Sun causes this process.

Matrix Matching Type

20. Correct Answer: a–P, b–Q, c–R, d–S
Water pollution means harmful substances in water (P). Air pollution involves release of harmful gases (Q). Precipitation is water falling from the sky (R). Distillation involves heating water to purify it (S).

21. Correct Answer: a–R, b–P, c–S, d–P
Troposphere is closest to Earth (R), as it is the lowest layer. Evaporation is process of water turning into gas (P). Filtration separates particles from water (S). Boiling water removes harmful bacteria (P).

Comprehension Type

Passage – I

22. Correct Answer: C
More than 70% of Earth's surface is covered by water, mainly in oceans. This makes Earth appear blue from space.

23. Correct Answer: B
Only about 3% of Earth's total water is freshwater. Most of the remaining water is salty and found in oceans.

24. Correct Answer: B
Industrial waste contains harmful chemicals that pollute water bodies. It is a major cause of water pollution.

Passage – II

25. Correct Answer: C
Weather occurs in the troposphere, the lowest layer of the atmosphere. Clouds, rain, and wind are found here.

26. Correct Answer: B
The ozone layer is present in the stratosphere. It protects Earth by absorbing harmful ultraviolet (UV) radiation.

27. Correct Answer: D
The exosphere is the outermost layer of the atmosphere. It gradually merges into outer space.

Previous Year Questions

1. **Correct Answer: B**
   When soil is heated, the water present in it evaporates and then condenses on the lid as droplets. This shows that soil contains water.
2. **Correct Answer: D**
   During the day, land heats up faster than the sea, so the air above land becomes warmer. The cooler air from the sea moves toward land, creating a sea breeze.
3. **Correct Answer: C**
   When the burning candle is covered with a glass, the flame goes out after some time. This happens because the oxygen inside is used up, showing that air (oxygen) is needed for burning.
4. **Correct Answer: B**
   Foggy breath in winter is due to condensation, where water vapour cools and forms tiny droplets. In the diagram, this process is represented by R.
5. **Correct Answer: C**
   Nitrogen is the most abundant gas in air, followed by oxygen, and then carbon dioxide in a very small amount. Hence, the correct decreasing order is Nitrogen > Oxygen > Carbon dioxide.

9. Earth and Universe

Worksheet – 1

1. **Answer: A) Sun**
   The Sun is at the center of our Solar System, and all the planets revolve around it.
2. **Answer: B) Mars**
   Mars is called the “Red Planet” because its surface contains iron oxide (rust), which gives it a reddish appearance.
3. **Answer: A) Saturn**
   Saturn is famous for its large and beautiful rings made mostly of ice particles and rocky debris.
4. **Answer: B) Jupiter**
   Jupiter is the largest planet in the Solar System, with a diameter of about 143,000 km.
5. **Answer: B) 8**
   There are 8 planets in the Solar System: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune.
6. **Answer: C) Jupiter**
   The Great Red Spot is a giant storm on Jupiter that has been raging for hundreds of years.
7. **Answer: D) Mercury**
   Mercury is the closest planet to the Sun.
8. **Answer: C) The Moon**
   The Moon is Earth's only natural satellite and orbits around Earth.
9. **Answer: C) Uranus**
   Uranus is tilted about 98°, causing it to rotate on its side and appear to roll around the Sun.
10. **Answer: B) Venus**
    Venus is often called Earth’s “twin” because it is very similar to Earth in size and mass.

Worksheet – 2

1. **Answer: B) Earth’s rotation on its axis**
   Day and night occur because Earth spins on its axis. The side facing the Sun experiences day, while the side facing away experiences night.
2. **Answer: B) 24 hours**
   Earth takes about 24 hours to complete one full rotation on its axis, creating one day-night cycle.
3. **Answer: B) Orbit**
   The path Earth follows around the Sun is called its **orbit**. One complete orbit takes about 365¼ days.
4. **Answer: C) Earth’s tilt on its axis and its revolution around the Sun**
   Seasons change because Earth is tilted about 23.5° and revolves around the Sun. Different parts of Earth receive varying amounts of sunlight throughout the year.
5. **Answer: C) Full Moon**
   During a Full Moon, the entire side of the Moon facing Earth is illuminated by the Sun, making it appear completely visible.

6. **Answer: A) New Moon**
   During a New Moon, the side of the Moon facing Earth is not illuminated, so it is generally not visible.
7. **Answer: C) Full Moon**
   A lunar eclipse can occur only during a Full Moon, when Earth is positioned between the Sun and the Moon, causing Earth's shadow to fall on the Moon.
8. **Answer: B) Only a small part of the Moon is visible**
   In the waxing crescent phase, a small illuminated portion of the Moon is visible, and the lit area increases each night.
9. **Answer: A) 29.5 days**
   The Moon takes about 29.5 days to complete one full cycle of phases, known as a lunar month.
10. **Answer: B) Axis**
    The axis is the imaginary line running through Earth's North and South Poles about which Earth rotates.
11. **Answer: B) Capricorn**
    Makar Sankranti is celebrated when the Sun enters Makara (Capricorn) according to the Hindu solar calendar.

Competitive Corner

Single Correct Option Type

1. **Answer: B) When the Moon blocks sunlight from reaching Earth**
   A solar eclipse occurs when the Moon passes between the Sun and Earth, blocking some or all of the Sun’s light from reaching Earth.
2. **Answer: B) 23.5°**
   Earth’s axis is tilted about 23.5°, which causes the changing seasons.
3. **Answer: B) summer**
   When the Northern Hemisphere tilts toward the Sun, it receives more direct sunlight and longer days, resulting in summer.
4. **Answer: B) both hemispheres receive nearly equal sunlight**
   Spring and autumn occur during the equinoxes, when both hemispheres get nearly equal amounts of sunlight.
5. **Answer: B) crescent moon**
   Just after a New Moon, a small curved illuminated portion becomes visible, called a crescent moon.
6. **Answer: B) When one celestial object blocks another**
   An eclipse happens when one celestial body moves into the shadow of another or blocks its light.
7. **Answer: A) Venus**
   Venus is the hottest planet because its thick atmosphere traps heat through a strong greenhouse effect.
8. **Answer: A) Saturn**
   Saturn is famous for its large and bright ring system, although other giant planets also have rings.
9. **Answer: B) ice, dust, and rock**
   Comets are often called “dirty snowballs” because they are mainly composed of ice mixed with dust and rocky material.
10. **Answer: B) Moons**
    Many planets have natural satellites called moons that orbit around them.

ONE OR MORE THAN CORRECT OPTION TYPE

11. Answer: A, C

The Moon orbits Earth and has almost no atmosphere. It is smaller than Earth and does not have weather like Earth.

12. Answer: A, C, D

The Sun is a star, the largest object in the Solar System, and its gravity keeps the planets in orbit. It is not made mostly of water.

13. Answer: A, B, C, D

Temperature, humidity, wind, and air pressure all affect weather conditions.

14. Answer: A, B, D

Weather describes the condition of the atmosphere, changes frequently, and can be rainy or windy. It is not the same everywhere

Assertion and Reason Type

15. Answer: A

Weather differs from place to place because factors like temperature, humidity, wind, and rainfall vary. Hence, both A and R are true, and R correctly explains A.

16. Answer: A

Earth's 23.5° axial tilt, along with its revolution around the Sun, causes seasons. Thus, R correctly explains A.

Matching Type Questions

17. Answer: A–R, B–S, C–Q, D–P

- Mars → The Red Planet
- Venus → Hottest planet with a toxic atmosphere
- Saturn → Planet with rings
- Jupiter → Largest planet with the Great Red Spot

18. Answer: A–Q, B–S, C–R, D–P

- The Sun → Centre of the Solar System
- The Moon → Natural satellite of Earth
- The Earth → Our home planet
- The Asteroid Belt → Region of rocky objects between Mars and Jupiter

Comprehension Type Questions

Passage-I

19. Answer: B

The Moon's gravitational pull causes tides in Earth's oceans.

20. Answer: C

Craters on the Moon were formed by meteorite impacts.

21. Answer: C

The Moon takes about 29.5 days to complete one orbit around Earth.

Passage-II

22. Answer: B

The Sun provides light and heat, which support life on Earth.

23. Answer: C

The Sun is mainly a giant ball of gas.

24. Answer: B

The Sun releases energy through the process of nuclear fusion.

Previous Year questions

1. **Answer:** C
   18th March is 6 days after the New Moon (12th March), appearing as a First Quarter moon. 21st March is 9 days after, appearing as a Waxing Gibbous.
2. **Answer:** D
   When the Moon is directly between the Sun and Earth, its illuminated side faces away from us, resulting in a completely dark New Moon.
3. **Answer:** D
   On 15th January, the Moon is a Waning Gibbous. Exactly seven days later on 22nd January, it continues its cycle and shrinks into a Waning Crescent.
4. **Answer:** C
   The Earth takes **1** day to complete a rotation on its axis (causing day and night) and **365** days to complete a revolution around the Sun (one year).
5. **Answer:** C
   5th October is 17 days after the New Moon (18th September), which corresponds to the Waning Gibbous phase. By 12th October (24 days later), it shrinks further into a Waning Crescent.

10. Our Environment

Worksheet – 1

1. **Correct Answer: C**
   Heavy rains after hot periods are typical of the monsoon season, which brings rainfall and makes fields green again.
2. **Correct Answer: A**
   Weather changes mainly due to the Sun heating the Earth unevenly and the movement of air.
3. **Correct Answer: C**
   Summer is a season, not a condition of weather, while hot, sunny, and cloudy describe weather.
4. **Correct Answer: B**
   Humidity refers to how much water vapour is present in the air.
5. **Correct Answer: A**
   When humidity is high, the air already contains a lot of moisture, so sweat evaporates slowly.
6. **Correct Answer: B**
   Smog is formed by the combination of smoke and fog.
7. **Correct Answer: C**
   Leaves fall (autumn), flowers bloom (spring), longest days (summer), and longest nights (winter).
8. **Correct Answer: B**
   Smog forms when smoke mixes with fog, often due to pollution from burning materials.
9. **Correct Answer: B**
   Smog pollutes the air and can lead to respiratory issues.
10. **Correct Answer: B**
    Trees help clean the air and reduce pollution, thereby reducing smog.

Worksheet – 2

1. **Correct Answer: B**
   Soil formation occurs through weathering, where rocks break down into smaller particles over time.
2. **Correct Answer: C**
   Loamy soil has a balanced mixture of sand, silt, and clay, making it ideal for plant growth and agriculture.
3. **Correct Answer: B**
   Overgrazing removes vegetation cover, leaving soil exposed to wind and water, causing erosion.
4. **Correct Answer: B**
   Soil pollution reduces soil quality and fertility, negatively affecting plant growth.
5. **Correct Answer: C**
   Cover crops protect the soil surface and reduce erosion by holding the soil in place.
6. **Correct Answer: B**
   Soil erosion is the removal of topsoil by natural forces like wind and water.
7. **Correct Answer: C**
   Topsoil contains the most organic matter (humus) and nutrients, making it the richest layer.

8. **Correct Answer: B**
Sandy soil has large particles and drains quickly, while clay soil holds water and drains slowly.

9. **Correct Answer: B**
Subsoil lies beneath the topsoil layer in the soil profile.

10. **Correct Answer: C**
Proper waste management helps prevent soil pollution and protects soil health.

Worksheet – 3

1. **Correct Answer: B**
National parks and wildlife sanctuaries are created mainly to conserve wildlife and protect their natural habitats.

2. **Correct Answer: B**
Biodiversity refers to the wide range of living organisms present in an ecosystem.

3. **Correct Answer: C**
Deforestation occurs when trees are cut down for wood, land use, or other human activities

4. **Correct Answer: B**
Conservation means protecting and managing natural resources so they remain available for future generations.

5. **Correct Answer: B**
Trees absorb carbon dioxide, so fewer trees mean less CO_2 is removed from the atmosphere.

6. **Correct Answer: B**
Protecting forests helps conserve wildlife because trees offer shelter and food for many species.

7. **Correct Answer: A**
Forests and oceans act as major carbon sinks, absorbing carbon dioxide and regulating climate.

8. **Correct Answer: A**
Climate change alters temperature and conditions, making habitats unsuitable for some species.

9. **Correct Answer: A**
Over-exploitation refers to using natural resources faster than they can be replenished.

10. **Correct Answer: A**
Minimizing waste lowers pollution and helps protect the environment.

Competitive Corner

Single Option Correct Type

1. **Correct Answer: C**
Weather changes mainly occur due to differences in temperature and air pressure, which cause winds and other weather conditions.

2. **Correct Answer: C**
A hygrometer is used to measure the amount of moisture (humidity) in the air.

3. **Correct Answer: D**
Winter has the shortest days and longest nights due to the tilt of the Earth away from the Sun.

4. Correct Answer: A
Topsoil is rich in organic matter, subsoil has more minerals, and bedrock is the solid rock layer beneath.

5. Correct Answer: B
Statements (i), (ii), and (iii) are correct. Statement (iv) is incorrect because the Sun shines more strongly in summer.

6. Correct Answer: B
Seasons occur because the Earth is tilted on its axis, causing different parts to receive varying sunlight.

7. Correct Answer: B
Protecting forest areas and stopping deforestation helps conserve wildlife by preserving their habitats.

8. Correct Answer: A
The movement of air is called wind.

One or More than One Correct Option Type:

9. Correct Answer: A, B, C
Summer, Winter, and Autumn are seasons. Cloudy describes weather, not a season.

10. Correct Answer: A, B, C
Deforestation, pollution, and over-exploitation harm the environment and living organisms. Photosynthesis is a natural beneficial process.

11. Correct Answer: A, B, D
Sandy, clay, and loamy are real soil types. Plastic soil is not a natural soil type.

Assertion and Reason Type

12. Correct Answer: A
Both the assertion and reason are true. Trees hold the soil with their roots, so when trees are removed (deforestation), soil becomes loose and erosion increases—this directly explains the assertion.

13. Correct Answer: A
Both the assertion and reason are true. Weather impacts farming, health, and safety, so understanding it helps people prepare and plan, which clearly explains why it is important.

Matrix Matching Type

14. Correct Answer: a–Q, b–S, c–R, d–P
Weather is the condition of the atmosphere at a particular time and place. Humidity is the amount of water vapour present in the air. Soil erosion is the wearing away of topsoil by natural forces like wind and water. Conservation means protecting and preserving natural resources.

15. Correct Answer: a–Q, b–P, c–R, d–S
Sandy soil has large particles and allows quick drainage. Monsoon is a season known for heavy rainfall. Compost is organic matter added to soil to improve fertility. These are species that are at risk of extinction.

Comprehension Type

Passage – I

16. Correct Answer: B
During the monsoon, there is a lot of water vapour in the air, which increases humidity.

17. Correct Answer: B
Monsoon humidity is associated with rainfall, which provides sufficient water for plant growth.

18. Correct Answer: B
Excess moisture during monsoon can lead to the growth of mold on clothes and walls.

Passage – II

19. Correct Answer: A
Wind and water are the main natural forces that carry away topsoil, causing soil erosion.

20. Correct Answer: B
Afforestation helps prevent erosion because tree roots hold the soil firmly in place.

21. Correct Answer: B
Soil erosion removes the fertile topsoil, leading to loss of fertile land.

Previous Year Questions

1. Correct Answer: C
(i) T – CNG is a cleaner fuel than petrol, so it is preferred.
(ii) F – Biodegradable wastes *can* be broken down by microorganisms.
(iii) F – Afforestation reduces soil erosion, not increases it.

2. Correct Answer: B
Water droplets form on the lid because soil contains moisture that evaporates and then condenses, showing the presence of water in soil.

3. Correct Answer: A
Factories and vehicles are decreasing over time, which reduces pollution sources, while the number of trees is increasing, which helps absorb pollutants. Together, these factors lead to a steady decrease in pollution over the years, as shown in option (A).

4. Correct Answer: B
Emission of pollutant gases from petrol and diesel vehicles contributes to both global warming (by releasing greenhouse gases like CO_2) and acid rain (by releasing gases like SO_2 and NO_x). Hence, it fits in the overlapping region X.

5. Correct Answer: A
Preventing domestic sewage from entering water bodies helps conserve clean water resources by reducing pollution and maintaining water quality.